01. Relativistic Time Dilation & Universal timeIn Architecture

Time Dilation DAW Technical Proof Series (v0.0.1)

1. Relativistic Coordinate Time & Dilation Rate

Coordinate time tau(t) is modulated by relativistic factor gamma(t) in [-16.0, 16.0]:

$$d\tau = \gamma(t) \cdot dt, \quad \tau(t) = \int_0^t \gamma(t')\, dt'$$

Special relativity velocity time dilation follows Lorentz factor:

2. Universal Node timeIn Propagation Architecture

Every node in Time Dilation DAW features an explicit or implicit Inlet 0 designated timeIn (NodePortType::Time).

Graph executes propagateTimeDilationHierarchy() prior to DSP processing:

$$\gamma_{\text{node}} = \gamma_{\text{patched}} \text{ if connected, else } \gamma_{\text{upstream}} \text{ if unpatched}$$

Sample-rate advance step d\tau_n = \gamma_{\text{node}}[n] / f_s modulates both sub-sample audio resampling and control phase accumulators.